

SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10

Sub-Strand 2.3: Trigonometry — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.3: Trigonometry
Driving Question	DRIVING QUESTION: How can we use the angles and one side of a right-angled triangle to find any unknown height or distance in our environment — without ever measuring it directly?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 The Water Tank Problem — Launching the Anchoring Phenomenon	Learners observed a scenario in which a raised water tank at a Kenyan school must have its vertical height determined, but direct measurement up the support structure is impractical. They measured or were given the horizontal ground distance from the base of the tank's support to their standing point, and used a clinometer (or a protractor-and-straw model) to record the angle of elevation to the base of the tank. Key observation: two pieces of data — one angle and one ground distance — are all that is available, yet a height must be found.	Learners identified that a right-angled triangle is embedded in the water-tank situation: the vertical height of the tank is the OPPOSITE side relative to the angle of elevation, the horizontal ground distance is the ADJACENT side, and the imaginary line of sight to the tank forms the HYPOTENUSE. They learned the vocabulary 'angle of elevation' and practised labelling the three sides of a right-angled triangle (opposite, adjacent, hypotenuse) relative to a named acute angle. They also articulated — in their own words on the Driving Question Board — that the relationship between the angle and the two sides must follow some fixed rule, which they will investigate in later lessons.	The teacher should use a large floor diagram (chalk on the courtyard, or a projected image of the school tank) to make the hidden triangle visible before any formula is introduced. Cold-call at least four learners to point out which line segment is the hypotenuse and which is the opposite side — WAIT 10 seconds after each prompt. Pedagogical note: resist giving the tangent formula at this stage; the purpose of Lesson 1 is to create productive uncertainty. The anchoring phenomenon (the tank problem) must remain 'open' so that every subsequent lesson feels like progress toward solving it. Record learners' initial predictions about the height on the DQB and revisit them in Lesson 8.
Lesson 2 Similar Right-Angled Triangles and Constant Ratios	Learners drew or measured a family of right-angled triangles that all share the same acute angle (e.g., 35°) but differ in size — mimicking the way the water-tank triangle would look from different distances. Using rulers, they recorded the lengths of the opposite and adjacent sides for each triangle. They computed the ratio $\text{opposite} \div \text{adjacent}$.	Learners discovered that the ratio $\text{opposite} \div \text{adjacent}$ remains constant across all similar right-angled triangles that share the same acute angle, regardless of the overall size of the triangle. This constancy is the foundational insight for trigonometry: the angle alone determines the ratio. Learners connected this back to the tank problem —	Instruct learners to work in pairs; each pair constructs at least three triangles with the same angle using graph paper or GeoGebra. After results are shared, ask: 'Why does the ratio stay the same even though the triangles are different sizes?' — WAIT 15 seconds, then cold-call three pairs. Pedagogical note: emphasise that

	adjacent for every triangle in the family and recorded results in a table.	if they know the angle and the ground distance, the constant ratio means they can calculate the unknown height. They also reviewed the properties of similar triangles (AA similarity) as the geometric justification for why the ratio is fixed.	this lesson provides the 'why' behind trigonometric ratios; learners who understand similar triangles will not need to memorise formulas blindly. Link explicitly to the DQB: 'We now know the ratio is constant — but what is the exact value for our tank's angle of 35°? That is our next question.'
Lesson 3 Deriving the Three Trigonometric Ratios: Sine, Cosine and Tangent	Learners measured all three sides of multiple right-angled triangles with fixed angles and computed all three possible ratios: opposite/hypotenuse, adjacent/hypotenuse, and opposite/adjacent. They then compared their computed values with values read from a trigonometric table (or a calculator) to confirm that the ratios have agreed-upon names and precise decimal values. The pattern observed: each ratio is constant for a given angle and each has a distinct name — sine, cosine, and tangent respectively.	Learners formally defined the three trigonometric ratios: $\sin \theta = \text{opposite/hypotenuse}$, $\cos \theta = \text{adjacent/hypotenuse}$, $\tan \theta = \text{opposite/adjacent}$. They used the mnemonic SOH-CAH-TOA (contextualised as 'Soma Chuo Hadi' — a Kenyan study-culture phrase — to aid memory). They also practised reading sine, cosine, and tangent values from trigonometric tables for angles between 0° and 90°, and verified these with a scientific calculator. They identified which ratio is most directly useful for the tank problem (tangent, since the known side is adjacent and the unknown side is opposite).	The teacher should derive each ratio on the board by naming it aloud as the class measures. After defining all three, pose: 'For the water-tank problem, we know the ground distance and the angle — which ratio connects those two pieces of information to the height?' — WAIT 10 seconds, cold-call two learners. Pedagogical note: do not rush past table-reading skills; many learners will use printed tables in examinations. Have learners cross-check at least five values between table and calculator so they trust both tools. Update the DQB: 'We now have a formula — $\tan \theta = \text{height/distance}$. But can we solve it yet?'
Lesson 4 Applying Trigonometric Ratios to Find Unknown Sides	Learners were presented with a series of right-angled triangle diagrams — drawn in Kenyan contexts (a mast at Kericho Tea Farm, a ramp at Mombasa Port, a footbridge over a stream in Kisumu) — in which one side and one acute angle are known and a second side is unknown. They observed how substituting values into $\sin \theta$, $\cos \theta$, or $\tan \theta$ and then rearranging the equation yields the unknown side.	Learners mastered the three algebraic rearrangements: $\text{opposite} = \text{hypotenuse} \times \sin \theta$; $\text{adjacent} = \text{hypotenuse} \times \cos \theta$; $\text{opposite} = \text{adjacent} \times \tan \theta$ (and their inverses). They practised selecting the correct ratio by first labelling all three sides relative to the given angle, then identifying which ratio links the known side to the unknown side. Learners solved at least six problems independently, including one that directly mirrors the water-tank scenario (given ground distance and angle of elevation, find height).	Use a structured problem-solving protocol: (1) Draw and label the triangle. (2) Name the known and unknown sides. (3) Choose the ratio. (4) Substitute and solve. (5) Check units and reasonableness. Cold-call learners at each step for at least two worked examples — WAIT 10 seconds at step 3 ('Which ratio?') since that is the highest-cognitive-demand moment. Pedagogical note: learners frequently confuse which side is opposite vs adjacent when the triangle is rotated; provide triangles in multiple orientations. Connect to DQB: the tank problem can now be solved numerically — build suspense for Lesson 8 where they will compute the actual answer.
Lesson 5 Sine and Cosine of Complementary Angles	Learners investigated pairs of complementary angles (angles that sum to 90°) within the same right-angled triangle. They measured or computed $\sin 30^\circ$ and $\cos 60^\circ$, $\sin 40^\circ$ and $\cos 50^\circ$, and $\sin \theta$ and	Learners proved — using the definitions of sine and cosine and the labelling of sides — that in any right-angled triangle, the sine of one acute angle equals the cosine of the other acute angle, because the side that is	Begin by asking learners to compute $\sin 55^\circ$ and $\cos 35^\circ$ on their calculators before revealing any rule — WAIT 20 seconds for the surprise of equal values to land. Then cold-call: 'Why are these the same number?'

	$\cos(90^\circ - \theta)$ for several values of θ. The observation: each pair of values is identical — $\sin \theta = \cos(90^\circ - \theta)$ for every angle tested.	'opposite' to one angle is 'adjacent' to the complementary angle. They stated the identity $\sin \theta = \cos(90^\circ - \theta)$ and its equivalent $\cos \theta = \sin(90^\circ - \theta)$. They practised using this relationship to simplify expressions and to find unknown angles when given a trigonometric equation (e.g., $\sin x = \cos 28^\circ \implies x = 62^\circ$).	What do you notice about 55° and 35°?' (WAIT 15 seconds). Pedagogical note: this lesson is frequently underweighted in teaching but is high-yield for KCSE questions. The complementary-angle identity also sets up Lesson 6's deeper structural insight. Ensure learners write the identity in their own words as well as in algebraic notation. DQB note: ask learners whether this identity could help them find the angle at the top of the water-tank triangle if they already know the angle at the ground.
Lesson 6 Sines and Cosines of Complementary Angles — The Hidden Shortcut in the Tank Problem	Learners returned to the water-tank right-angled triangle and labelled both acute angles: the angle of elevation θ at the ground, and the complementary angle ($90^\circ - \theta$) at the top of the triangle. They then wrote expressions for $\sin$ and $\cos$ of both angles and observed that the side lengths referenced swap roles depending on which angle is chosen as the reference angle.	Learners deepened their understanding of the complementary-angle identity by seeing it structurally: the 'opposite' side of the lower angle is the 'adjacent' side of the upper angle, and vice versa, because the two acute angles are complementary. They applied this insight to re-express the tank height using both $\sin$ and $\cos$ — arriving at the same numerical answer via two different ratio choices — confirming that the complementary identity is not merely algebraic but geometrically inevitable. They also practised mixed problems that require choosing between the two forms of the identity strategically.	Instruct learners to colour-code the sides in the tank diagram: one colour for the side opposite the lower angle, another for the side opposite the upper angle, and note that the colours swap roles. Ask: 'If I stand closer to the tank, does the complementary relationship between the two angles change?' — WAIT 15 seconds; cold-call two learners. Pedagogical note: this lesson distinguishes surface memorisers from deep understanders; use targeted questioning to reveal misconceptions. Confirm on the DQB that the two acute angles always sum to 90° regardless of the tank's height or the observer's distance, reinforcing the structural — not accidental — nature of the identity.
Lesson 7 Clinometer Construction & Field Calibration — Building the Measurement Tool	Learners constructed simple clinometers (protractor + straw + weighted string) and used them in the school compound to measure angles of elevation to the top of a classroom wall, a flagpole, or a nearby tree in Nairobi's typical school setting. They recorded (ground distance, angle of elevation) pairs from two different positions and computed the height both times using $h = d \times \tan \theta$. They observed whether both calculations yield the same height, thereby calibrating the accuracy of their instrument.	Learners applied the formula $h = d \times \tan \theta$ in a real, outdoor measurement context and understood that a consistent height from two different (d, θ) pairs validates both the formula and the measurement tool. They identified sources of error (parallax in reading the clinometer, uneven ground, string not hanging freely) and discussed how to minimise them. They also practised rounding answers to an appropriate degree of precision and reporting their results with correct units (metres).	Assign roles within each group (angle reader, distance measurer, recorder, calculator) and rotate roles between the two measurement positions so every learner practises each skill. After groups report their two height values, cold-call: 'Your two answers differ by 0.4 m — is that acceptable? What could cause this?' — WAIT 20 seconds. Pedagogical note: the field activity is the experiential bridge to Lesson 8's final solution of the tank problem; treat accuracy and error discussion as mathematically important, not merely procedural. Groups should photograph their setup and triangle diagram for inclusion in the DQB gallery.

Lesson 8 Final Explanation: Solving the Water-Tank Problem and Closing the Explanatory Loop	Learners revisited all data collected or given in Lesson 1 — the horizontal ground distance from the tank support and the angle of elevation to the base of the tank — and applied everything learned across the unit. They observed their own DQB entries from Lesson 1 and compared their initial (often intuitive) predictions of the height with the now-calculated answer.	Learners demonstrated that $\tan \theta = \text{opposite/adjacent} = \text{height/horizontal distance}$ can be rearranged to $h = d \times \tan \theta$, and they computed the actual numerical height of the water tank. They generalised the method: any unknown height or distance in the environment can be found if one angle and one side of the resulting right-angled triangle are known. Learners also reflected on which trigonometric ratio to use in different scenarios (sine when the hypotenuse is known, cosine when the adjacent and hypotenuse are involved, tangent when opposite and adjacent are the relevant pair) and articulated a decision procedure for future problems.	Open by asking learners to read their Lesson 1 DQB predictions aloud before revealing the calculated answer — this closing of the loop is pedagogically powerful and should not be rushed. Cold-call at least three learners to narrate the full solution in their own words before the teacher writes anything on the board. Then pose a transfer question: 'A mobile-phone mast in Rift Valley casts a shadow 18 m long when the sun's angle of elevation is 72°. How tall is the mast?' — WAIT 20 seconds, then cold-call. Pedagogical note: the final lesson should feel like a celebration of accumulated understanding, not a new lesson. Ensure the DQB is fully annotated and that every earlier entry has been 'answered' by a subsequent lesson. Award recognition to the most improved DQB group.
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